



University of the Philippines Manila
College of Arts and Sciences



CERTIFICATE OF RECOGNITION

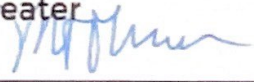
This certificate is presented to:

Ma. Adela E. Arenas

For winning the **Dean's Award for Outstanding Research** in Physical & Computing Sciences category during the "**1st CAS Undergraduate Research Colloquium**" held on the 18th day of June, 2024 at the UP Manila College of Arts and Sciences.

Given during the 2024 CAS Recognition Program, this 19th day of July 2024 at the University of the Philippines Diliman Theater


Christopher Jay T. Robidillo, RCh, MSc, PhD
Associate Dean for Research and Public Service


Maria Constanca O. Carrillo, PhD
Dean, CAS UP Manila